

ASR Final Report

Project goal

The project evaluates the Hebrew ASR model ivrit-ai/whisper-large-v3-turbo-ct2 on the Mozilla Common Voice Hebrew test set.

Part A

For Part A, I transcribed the test.tsv split and saved the results in:

- `results\_part\_a.tsv`

The output format is:

- `Filename`
- `Reference Text`
- `Transcribed Text`

All text files were handled as UTF-8.

Part B: baseline evaluation

The baseline evaluation was computed from the raw transcription output using word-level alignment and edit statistics.

- Input file: `results\_part\_a.tsv`
- Output file: `results\_part\_b.tsv`
- Alignment log: `part\_b\_alignment\_log.tsv`
- Frequent errors report: `part\_b\_frequent\_errors.tsv`

Final baseline results on the clean benchmark:

- WER: `0.3592`
- Recall: `0.6578`
- Precision: `0.6554`
- F1: `0.6566`

Part C: error analysis and normalization

Analysis method

I analyzed the model output in two complementary ways:

1. I inspected the full alignment log for individual utterances.
2. I reviewed the most frequent error pairs over the full benchmark.

This made it possible to separate true ASR errors from artifacts caused by the evaluation procedure.

Main issues found

The raw evaluation penalized the system for many differences that do not reflect real recognition failures:

- punctuation differences
- Hebrew vowel marks, diacritics, and quote marks
- spelling variants that keep the same meaning
- orthographic variants of common Hebrew words
- some date, number, and percentage expressions that appeared in equivalent surface forms

The goal was to change only surface form differences and same-meaning variants, as required by the assignment, and not to rewrite genuinely different content.

Final normalization rules

The final normalization pipeline performs:

1. removal of Hebrew vowel marks and other diacritics
2. removal of punctuation and quote-like symbols
3. whitespace normalization
4. normalization of common Hebrew word variants that preserve the same meaning
5. normalization of selected date, number, and percentage forms used in equivalent contexts

Iterative improvement

The normalization was developed in stages and evaluated after each stage on the full benchmark:

1. Raw baseline evaluation: `WER 0.3592`
2. Conservative normalization of vowel marks and punctuation: WER dropped substantially relative to the raw baseline
3. Expanded normalization of same-meaning orthographic variants and selected numeric/date forms: `WER 0.0567`

Final clean-benchmark results

- Input file: `results\_part\_a.tsv`
- Output file: `results\_part\_c.tsv`

- Alignment log: `part\_c\_alignment\_log.tsv`
- Frequent errors report: `part\_c\_frequent\_errors.tsv`

Final normalized results on the clean benchmark:

- WER: `0.0567`
- Recall: `0.9465`
- Precision: `0.9475`
- F1: `0.9470`

The official submitted benchmark result is the TOTAL row in results_part_c.tsv.

This reached the assignment target of single-digit WER.

Part D: noisy audio robustness

For the noise condition, the assignment requires the mod 6 = 0 setting:

- background type: noise
- SNR range: `0-6 dB`

Only long files from MUSAN/noise/free-sound/ were used, as required. Each Common Voice test clip was downsampled to 16 kHz using `scipy.signal.decimate(..., q=2)` before mixing.

Generated files:

- augmentation log: `augmentation\_log.tsv`
- noisy transcription file: `results\_part\_a\_noisy.tsv`
- noisy normalized evaluation file: `results\_part\_c\_noisy.tsv`
- noisy alignment log: `part\_c\_noisy\_alignment\_log.tsv`
- noisy frequent errors report: `part\_c\_noisy\_frequent\_errors.tsv`

Final normalized results on the noisy benchmark:

- WER: `0.1682`
- Recall: `0.8437`
- Precision: `0.8474`
- F1: `0.8456`

The official submitted noisy result is the TOTAL row in results_part_c_noisy.tsv.

Interpretation

The model remains usable under strong background noise, but performance drops substantially relative to the clean benchmark:

- clean normalized WER: `0.0567`
- noisy normalized WER: `0.1682`

The increase in WER shows that background noise causes more substitutions, deletions, and insertions, especially function words and short words.

Conclusion

The raw baseline significantly underestimated the real quality of the model because it penalized many formatting and orthographic differences. After targeted normalization based on same-meaning word variants, removal of vowel marks, removal of punctuation, and selected equivalent number/date forms, the clean-benchmark WER dropped from 0.3592 to 0.0567. Under strong background noise, the normalized WER increased to 0.1682, showing a clear but expected robustness degradation.